[BENG - 1230]

I/ IV B.Tech. DEGREE EXAMINATION

Second Semester

PHYSICS

(Group - A : Common for branches: Civil, CSE & IT Chemical, and Biotechnology)

(Effective from the admitted batch of 2020-2021)

Time : 3 Hours   Max. Marks: 70

Question No.1 is compulsory.
Answer any FOUR from the remaining questions .
All questions carry equal marks.

1. (a) What is entropy? Give its physical significance. (4)

   (b) How are the magnetic materials classified? (3)

   (c) Show that at Brewster angle, the reflected ray is perpendicular to refracted ray. (4)

   (d) Explain Heisenberg's uncertainty principle. (3)

2. (a) State and explain first law in thermodynamics. (6)

   (b) State and prove Carnot's theorem. (8)

3. (a) State Gause's law in electrostatics. Calculate the electric field due to an infinite conducting sheet of charge. (7)

   (b) State Ampere's circuital law. Explain its application to solenoid carrying a current. (7)

(P.T.O.)

(a) What are ultrasonic waves? Write their properties. (4)

(b) Describe how ultrasonic waves are produced by magnetostriction method with diagram. Write any five applications. (10)

(a) What is interference of light? Describe Young's experiment. (7)

(b) What do you understand by double refraction? Explain Huygens's theory of double refraction in uniaxial crystals. (7)

(a) Describe the construction and working of gas laser. (8)

(b) How do you classify optical tibres? (6)

. (a) Obtain Schrodinger wave equation for a particle in a box. (8)

(b) Write a notes on free electron theory of metals. (6)

3. (a) What are nano phase materials? (4)

(b) Write any two methods for the synthesis of nano materials. Give any few applications of nano materials. (10)

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